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lec#

①

★

i) $\int 2x(x^2+1)^{23} dx$

Sol

$$= \int (x^2+1)^{23} \cdot 2x dx$$

$$= \boxed{\frac{(x^2+1)^{24}}{24} + C} \quad \text{Ans}$$

ii) $\int \cos^3 x \sin x dx$

Sol

$$= \int (\cos x)^3 \cdot (\sin x) dx$$

$$= \boxed{-\frac{\cos^4 x}{4} + C} \quad \text{Ans}$$

②

i) $\int \frac{1}{\sqrt{x}} \sin \sqrt{x} dx$

Sol

$$= -2 \int \frac{1}{2\sqrt{x}} \cdot -\sin \sqrt{x} \cdot dx$$

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$$= \boxed{-2 \cos \sqrt{x} + C} \quad \text{Ans}$$

$$\text{ii) } \int \frac{3x dx}{\sqrt{4x^2 + 5}}$$

Sol, let

$$u = 4x^2 + 5$$

$$\frac{du}{dx} = 8x$$

$$du = 8x dx$$

$$dx = \frac{du}{8x}$$

Thus

$$= \int \frac{3x}{\sqrt{\frac{u}{4}}} \cdot \frac{du}{8x}$$

$$= \frac{3}{8} \int \frac{1}{\sqrt{u}} du$$

$$= \frac{3}{8} \left[\frac{u^{1/2}}{1/2} \right]$$

$$= \frac{3}{8} 2\sqrt{2x}$$

$$= \boxed{\frac{3}{4} \sqrt{4x^2 + 5} + C} \quad \text{Ans}$$

i) $\int \sec^2(4x+1) dx$ (3)
Sol

$$= \boxed{\frac{\tan(4x+1)}{4} + C}$$

i) $\int \sqrt{\sin x} \cos x dx$ (4)
Sol

$$= \boxed{\frac{2\sqrt[3]{\sin x}}{3x} + C} \quad \text{Ans}$$

$$\boxed{\frac{2(\sin^{3/2}(x))}{3x} + C} \quad \text{Ans}$$

i) $\int \cot x \csc^2 x dx$ (5)

$$= - \int (\cot x - \csc^2 x) dx$$

$$= \cancel{-\cot x + C} \quad \text{Ans}$$

$$= \boxed{-\frac{\cot^2 x + C}{2}}$$

i) $\int \cos 2x dx$ ①

Sol

$$= \frac{2}{2} \int \cos 2x dx$$

$$= \frac{1}{2} \int \cos 2x \cdot 2 \cdot dx$$

$$= \boxed{\frac{\sin 2x + C}{2}}$$

i) $\int x^2 \sqrt{1+x} dx$ ⑦

Sol, let

$$= u = 1+x, \quad x = u-1$$

$$= du = dx, \text{ Then}$$

$$= \int x^2 \sqrt{u} du$$

$$= \int (u-1)^2 \sqrt{u} du$$

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$$\begin{aligned}
 &= \int (u)^{1/2} (u^2 - 2u + 1) \\
 &= \int u^{5/2} - 2u^{3/2} + u^{1/2} \\
 &= \int u^{5/2} du - 2 \int u^{3/2} du + \int u^{1/2} du \\
 &= \frac{u^{7/2}}{7/2} - 2 \times \frac{u^{5/2}}{5/2} + \frac{u^{3/2}}{3/2} \\
 &= \frac{2}{7} u^{7/2} - \frac{4}{5} u^{5/2} + \frac{2}{3} u^{3/2} \\
 &= \left[\frac{2}{7} (1+x)^{7/2} - \frac{4}{5} (1+x)^{5/2} + \frac{2}{3} (1+x)^{3/2} + C \right] \quad \text{Ans}
 \end{aligned}$$

i) $\int \sin(x - \pi) dx$ ⑧

Sol

$$= \boxed{1 - \cos(x - \pi) + C}$$

i) $\int \frac{dx}{x \ln x}$ ⑨

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Sol

$$= \int \frac{1}{\ln x} \cdot \frac{1}{x} dx$$

$$= \boxed{\ln(\ln x) + C} \quad \text{Ans}$$